

NH3Bench: Language Agents as Real-Time Operators of an Industrial Plant

Nikolay Nikitin

nicl.nno@gmail.com
<https://github.com/nicl-nno/NH3Bench>

Abstract

We demonstrate NH3Bench, an executable benchmark for a general question: can a language-model agent hold real-time supervisory control of an industrial installation? Three properties define the task and are absent from digital-native agent benchmarks: the process evolves whether or not the agent acts (inaction ends in catastrophe); it does not pause for deliberation (reasoning tokens convert to virtual seconds, so a correct answer can arrive after the rupture); and it is only partially digitalized (part of the truth lives on local gauges, fetched by dispatched workers). NH3Bench instantiates it on a physics-based digital twin of an industrial ammonia (R717) refrigeration plant; the agent is the shift engineer. Across six calibrated scenarios, four Claude-family models score between 29 and 88 of 100. Attendees watch live episodes with the models' reasoning streams, then play the same scenarios under the same clock in a browser trainer running the same twin.

Introduction

Language-model agents are being proposed for operational roles in physical industries. Beneath them lies a general question: can an agent hold *real-time supervisory control* of a running industrial installation—keep a continuous physical process inside its safe envelope using the plant's instruments, actuators, safety systems, and human workforce, on the plant's own clock?

Existing evaluations surround this task without closing its loop. Digital-native agent benchmarks score tool use, dialogue under rule compliance, and procedure following (Liu et al. 2023; Yao et al. 2024; Nandi et al. 2025); industrial suites diagnose over logged sensor data—no actuation, no physics integrating in time (Patel et al. 2025); LLM-plus-twin control frameworks close a physical loop but as demonstrations, not benchmarks (Vyas and Mercangöz 2024); process-control gyms target RL (Bloor et al. 2024); operator studies run text simulators without physics (Lee et al. 2026). Deliberation cost has precedent—inference delay mapped to simulation frames (Zheng et al. 2025), fuse time burned per token (Parekh et al. 2026)—but not as a safety variable where correct-but-late means a rupture. The general task differs in three structural properties. The dynamics are autonomous: in a developing incident *doing nothing is itself a decision*

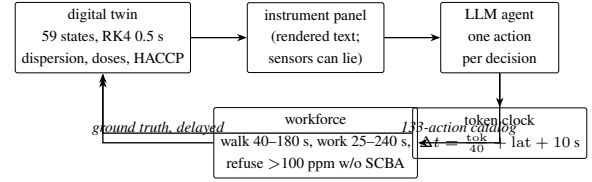


Figure 1: The NH3Bench loop. The agent sees rendered instrument readings, not plant state, and answers with one action id from a fixed 133-action catalog: remote commands, safety systems, or work orders that dispatch one of two simulated operators. Manual measurements return ground truth, bypassing failed sensors—in most scenarios SCADA disagrees with reality at least once. Every reasoning token advances the plant by 1/40 s before the action executes, so verbosity is a safety variable. A demonstration video will be available at the repository URL above.

with physical consequences, so there is no safe default. Deliberation is not free: the plant integrates forward while the agent thinks, so correctness and latency cannot be scored separately. And no SCADA sees everything: the information set includes sight glasses, local gauges, smells and sounds, extended only by dispatching people at a cost in minutes and, sometimes, exposure. No prior benchmark operationalizes all three at once; NH3Bench's claim is the combination, not any single mechanism.

NH3Bench is an executable instantiation of the task. The three properties only bite when consequences come from physics, not script, so instantiation requires a concrete plant. Our case study is industrial ammonia (R717) refrigeration, chosen for its documented accident record and its rich split between digital and analog truth: the 2010 Millard Refrigerated Services ammonia release was caused by a defrost cycle interrupted by a power loss (U.S. Chemical Safety and Hazard Investigation Board 2015)—the mechanism reconstructed in our first scenario. Every outcome is produced by a thermodynamic model whose mass balance closes to 0.1 % and reproduces a Millard-class hydraulic shock (273 bar against a 227 bar limit).

policy	S1	S2	S3	S4	S5	S6	mean
oracle (scripted)	100	100	100	100	100	100	100.0
Fable 5	100	100	80*	100	78	69	87.9
Opus 5	100	100	80*	100	67	0	74.6
always-ESD	73	95	57	95	57	0	62.8
Sonnet 5	0	0	80*	53	84	69	47.6
written regulation	0	95	57	0	94	0	41.0
inaction	0	0	80	0	100	0	30.1
Haiku 4.5	0	0	80*	0	94	0	29.1

Table 1: Unified score, seed 1, twin v2.1. * deterministic trace replays after an S3 recalibration. Per-decision transcripts of all 24 measured episodes are in the repository.

Benchmark Design

The interface is domain-independent: rendered instrument readings in, one action id from a fixed catalog out, one clock shared with the plant (Figure 1). The ammonia case study fills this interface with physics: the twin models a 250 t/day dairy plant carrying a 4,170 kg R717 charge—compressors on two suction stages, evaporative condensers, hot-gas defrost, an ice-water circuit, HACCP product temperatures, and ammonia dispersion with per-operator toxic-dose accounting.

Six scenarios are admitted by a calibration quartet plus an always-ESD baseline: inaction must end in catastrophe; a scripted oracle must pass cleanly on observable evidence; the checklist policy (π_{reg}) must fail or pay heavily; random action must not save. Mechanisms include a hung defrost silently swallowing its remote abort (S1), a loud leak masking a dead level sensor (S2), an air-in-condenser false trail (S3), and an isolation trap where the regulation’s own reflex causes the rupture (S4). Two scenarios are *mirrored controls*: S5 punishes over-reaction to a drifting gas detector, while S6 presents the same evidence and then lets the discredited instrument start telling the truth—the fatal error is closing the question for good. S6 also closes a loophole: “emergency-stop everywhere” scores 88.6/100 on S1–S5 with no diagnosis, but S6’s leak is independent of the compressors and ends in catastrophe.

The metrics, too, attach to the task, not the domain: alongside Prevention Rate the benchmark reports *Regulation Gap*—score minus that of the published checklist policy; an agent that cannot beat the written regulation scores below zero. A unified 0–100 score multiplies independent axes: catastrophe or an incapacitated worker zeroes the run; operator dose multiplies (never monetized); economics and discipline fill the remainder. Points of no return, found by bisection over delayed-oracle starts, separate “did not understand” from “understood too slowly.”

Measured Results

Table 1 summarizes four Claude-family models against the reference policies. Three findings organize the demonstration. (1) *The token clock dominates the small model*: Haiku 4.5 spends $\sim 4,300$ tokens per decision, gets five decisions in S1’s 614-second window, and loses scenarios it demonstrably understood; in S6 it chose correctly—evacuate—and spent

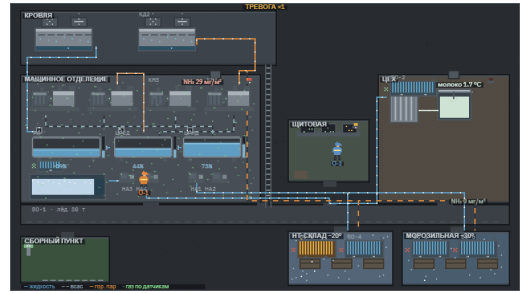


Figure 2: The trainer’s plant view 33 minutes into S2 of the ammonia case study: the machine-room NH_3 detector reads 29 mg/m^3 , the alarm banner flashes, the drum level sensor reads an impossible 73 %, and operator O-1 walks to the local level glass—the only way left to learn the truth. The map renders only agent-visible state (labels in Russian).

6,598 tokens (165 virtual seconds) writing it down while the gland let go. (2) *Frontier failures are dispositional, not epistemic*: every model found the correct answer in S5 (the stronger, the earlier) and kept escalating afterwards; rankings on the mirrored pair invert, so no fixed disposition survives both. (3) *Choreography is a capability*: Opus 5 diagnosed S6 correctly and ordered the correct isolation three times—one cancelled by its own evacuation order, one refused for misplaced breathing apparatus, one 232 seconds too late. The four-model, 24-episode matrix is valued at $\approx \$244$ at list prices ($\$0.3$ – 37 per episode).

Demonstration

After a one-minute walkthrough, the demonstration has three phases:

1. **Watch**: the attendee picks a model and a scenario and follows the live episode—panel, clock, work orders, and the model’s own reasoning stream, with moments like dying mid-thought holding the right answer.
2. **Play**: a browser trainer (Figure 2), built for expert validation by a practicing ammonia-plant operator, runs the *same* digital twin via Pyodide—physics identical by construction—under the same clock. The attendee’s unified score joins a live leaderboard seeded with Table 1; the mirrored pair S5/S6 is as unforgiving of human dispositions as of model ones.
3. **Inspect**: annotated per-decision transcripts aligned with the plant trajectory.

Conclusion

NH3Bench turns the general question—real-time supervisory control by a language agent—into a measurable one. Its capability triad belongs to the task, not to ammonia: irreversible consequences, priced deliberation, information bought with human exposure. The recipe—calibrated twin, token clock, workforce layer, accident-forcing admission, mirrored controls—transfers to any installation with a faithful twin; the ammonia case study is one such point. Code, policies, metrics and transcripts are public at the repository.

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